



OPEN Effect of multimodal preventive analgesia based on serratus anterior plane block and oxycodone on postoperative analgesia in elderly patients undergoing thoracoscopic lobectomy: a randomized controlled trial

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Patients undergoing thoracoscopic lobectomy often experience significant postoperative pain, which is frequently inadequately managed in elderly patients due to their unique physiological characteristics. Multimodal preventive analgesia has been shown to provide satisfactory pain relief and promote early recovery. The objective is to optimize perioperative pain management in elderly patients by multimodal preventive analgesia through a combination of serratus anterior plane block (SAPB) and oxycodone. A total of 80 elderly patients with lung cancer undergoing elective uniportal video-assisted thoracoscopic lobectomy under general anesthesia were enrolled, classified as American Society of Anesthesiologists (ASA) II or III, were randomly assigned to four groups. Group M1 received ultrasound-guided SAPB with 0.375% ropivacaine combined with intravenous oxycodone. Group M2 received intravenous oxycodone. Group M3 received SAPB, while the control group (Group C) was included for comparison. The primary outcome measures included the Visual Analogue Scale (VAS) scores at rest and during coughing immediately after postoperative tracheal extubation. Secondary outcome measures comprised VAS scores at rest and during coughing at 6 h, 24 h, 48 h, and 72 h postoperatively; intraoperative remifentanyl consumption; number of activations of the Patient-Controlled Intravenous Analgesia (PCIA) pump; frequency and dosage of rescue analgesia; intraoperative vasoactive drug consumption; and incidence of postoperative adverse events. The experimental groups had significantly lower VAS scores compared to the control group at all measured time points. In the M1 group, VAS scores at 6 h and 24 h after surgery were significantly lower than those in the other two experimental groups. The secondary outcomes are statistically significant differences among the four groups. Multimodal preventive analgesia based on SAPB and oxycodone can effectively reduce perioperative pain in elderly patients undergoing thoracoscopic lobectomy, decrease intraoperative and postoperative opioid consumption, and facilitate recovery.

Trial registration: ChiCTR2400088399.

Keywords Multimodal preventive analgesia, Serratus anterior plane block (SAPB), Oxycodone, Postoperative pain

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The global population is aging rapidly. In 2021, there were 761 million individuals aged 65 and older, a figure projected to reach 1.6 billion by 2050. Notably, the population aged 80 and above is expanding at an even faster rate. Anesthesiologists must prepare for a future in which the majority of patients will be over 65 years old, with a substantial proportion exceeding 80 years. Most studies suggest an increased pain threshold in older adults¹. However, Helme et al.^{2,3} reported that when the stimulus duration is short, the thresholds for heat- and electrically-induced pain increase in the elderly. In contrast, more intense or prolonged stimuli tend to result in higher pain reports, which may be more pronounced with advancing age. Combined with age-related physiological and pathological changes, comorbidities, and alterations in drug metabolism, pain management in elderly patients poses significant challenges⁴. Inadequate understanding of postoperative pain in elderly patients has led to poor pain control, with approximately 50–75% of older patients reporting insufficient pain relief after surgery⁵. Inadequate perioperative analgesia may suppress the body's immune function, increase the incidence of cardiovascular and cerebrovascular events, exacerbate cognitive and emotional disorders in elderly patients, delay postoperative recovery, and even lead to the development of chronic postoperative pain, affecting long-term prognosis and quality of life⁶.

Therefore, ensuring safe and precise pain management for elderly patients to alleviate acute postoperative pain and accelerate recovery is the focus of our attention. While multimodal analgesia and preventive analgesia are individually established approaches in pain management, their integration—referred to as multimodal preventive analgesia—has been proposed to maximize the benefits of both strategies.

Recent studies have explored multimodal preventive analgesia across various surgical contexts, yet findings remain inconsistent. Existing research predominantly focuses on middle-aged and younger populations undergoing abdominal, orthopedic, or oromaxillofacial procedures, with limited attention to thoracic surgeries^{7–9}.

Based on the above background, we conducted a study to examine the impact of SAPB combined with oxycodone in multimodal preventive analgesia on the analgesic effect in elderly patients undergoing thoracoscopic lobectomy procedures.

Methods

Study design and patients

This randomized, prospective, double-blind study was conducted from August 2024 to November 2024 after obtaining approval from the First Affiliated Hospital of Zhengzhou University Ethical Committee (2024-KY-0526-001). This study was prospectively registered with the Chinese Clinical Trial Registry on 19/08/2024 (ChiCTR2400088399). Written informed consent was obtained from all participants. Lung cancer patients undergoing elective uniportal thoracoscopic lobectomy under general anesthesia; aged 65 to 85 years old; classified as American Society of Anesthesiologists (ASA) physical status II/III; with a body mass index (BMI) between 18 and 32 kg/m² were eligible for inclusion. Exclusion criteria encompassed individuals with preoperative acute or chronic pain; a history of opioid therapy; severe respiratory, cardiovascular, or cerebrovascular diseases (hypertension controlled with SBP ≤ 160 mmHg and DBP ≤ 90 mmHg); severe liver or kidney dysfunction (AST/ALT levels exceeding 1.5 times the upper limit of normal, creatinine/urea nitrogen exceeding 1.2 times the upper limit of normal), local anesthetic allergy, opioid abuse, or communication disorders.

Patients were assigned to M1, 2, 3 or C groups according to the analgesia protocol. Randomization was performed using computer-generated random numbers. Sealed envelopes were used for allocation concealment. Participants were enrolled by clinical staff who were not involved in the allocation process. The allocation sequence was concealed and only revealed to the assigning investigator after participant enrollment. All procedures performed involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Intervention

Following the randomisation method, patients were randomly divided into four groups: M1 group (Ultrasound-guided SAPB with 0.375% ropivacaine 30 ml combined with intravenous infusion of oxycodone 0.1 mg/kg 3 min before skin incision), M2 group (intravenous infusion of normal oxycodone 0.1 mg/kg 3 min before skin incision), M3 group (Ultrasound-guided SAPB with 0.375% ropivacaine 30 ml) and C group. The anesthesiologists performing the nerve block procedure were blinded to the group allocation. Patient-controlled intravenous analgesia (PCIA) was used to manage pain after surgery. The formulation included hydromorphone (0.2 mg/kg), palonosetron (0.15 mg), and normal saline, diluted to a total volume of 200 ml. The background infusion rate was set to 2 ml/h, with a single bolus dose of 4 ml and a lockout interval of 15 min. If the postoperative VAS scores exceed 3, 3 mg of oxycodone was administered intravenously as rescue analgesia by an anesthesiologist who was blinded to the group assignments. Following surgery, the anesthesiologist checked in with the patients and recorded their pain levels and any adverse events, including nausea, vomiting, or respiratory depression.

After patients were recruited, hospital staff used an operating room monitor to monitor patient vital signs such as ECG, oxygen saturation (SpO₂) and blood pressure, with invasive blood pressure monitoring obtained by radial artery puncture. 4 L/min of oxygen were administered. SAPB was performed according to the technique described by Blanco et al.¹⁰ After establishing intravenous access, each patient was placed in a supine position. Following aseptic skin preparation with iodophor, the high-frequency linear array ultrasound probe was positioned over the midclavicular region of the thoracic cage in the sagittal plane. The medical ultrasound equipment employed in this study was manufactured by Fujifilm Holdings Corporation (Japan). Subsequently, the subcutaneous tissue, latissimus dorsi, serratus anterior, intercostal muscles, and the pleura superficial to the fourth and fifth ribs at the midaxillary line were identified. The deep SAPB was targeted to the interfascial plane between the serratus anterior and the rib periosteum. SAPB was conducted using an ultrasound-guided

in-plane approach with a 22 G needle. Approximately 20 min after SAPB, cold and pinprick tests along the midclavicular, midaxillary, and midscapular lines were performed. Successful sensory block was defined as a markedly reduced perception of cold (Fig. 1). This trial strictly adhered to the initial research protocol from design to implementation, with no changes made to any predefined outcome measures, data collection time points, or statistical analysis methods.

Anesthesia

Midazolam 2 mg, etomidate 0.2 mg/kg, fentanyl 50 µg/kg, and rocuronium 1 mg/kg were used to facilitate endotracheal intubation. All patients were intubated with a left-sided double-lumen endotracheal tube, and the position was confirmed using fiberoptic bronchoscopy to ensure effective bilateral lung isolation. A multifunctional monitoring system (HXD-I) was utilized to continuously track the Pain Threshold Index (PTi) and Wavelet Index (WLi). Both indices were maintained between 40 and 60 throughout the procedure. Propofol and remifentanyl infusion rates were dynamically adjusted to maintain both indices within the 40–60 range: the propofol infusion rate was increased by 1 mg kg⁻¹ h⁻¹ if WLi exceeded 60 for > 1 min and decreased by 1 mg kg⁻¹ h⁻¹ if WLi fell below 40 for > 1 min, while the remifentanyl infusion rate was adjusted by 2 µg kg⁻¹ h⁻¹ (increased for PTi > 60 or decreased for PTi < 40 sustained > 1 min), with intermittent rocuronium boluses administered as needed. Pressure-controlled ventilation (PCV) was implemented with a driving pressure < 25 cmH₂O, PEEP of 5 cmH₂O, and tidal volumes maintained at 6–8 mL/kg during dual-lung ventilation and 5–6 mL/kg during one-lung ventilation. The initial inspired oxygen concentration (FiO₂) during one-lung ventilation was set at 0.4, which was increased incrementally by 0.1 if SpO₂ dropped below 92%. Permissive hypercapnia was allowed under predefined thresholds (PaCO₂ < 65 mmHg, pH > 7.20).

Outcome measurements

The primary outcome was VAS scores (0–10, 0 = no pain, 10 = worst pain imaginable) at rest and during coughing immediately after postoperative tracheal extubation.

The secondary outcomes included VAS scores at rest and during coughing at 6 h, 24 h, 48 h, and 72 h postoperatively; intraoperative remifentanyl consumption; number of activations of the Patient-Controlled Intravenous Analgesia (PCIA) pump; frequency and dosage of rescue analgesia; intraoperative vasoactive drug consumption; and incidence of postoperative adverse events.

Heart rate (HR), mean arterial pressure (MAP), Wavelet Index (WLi), and Pain Threshold Index (PTi) were recorded and compared across all four groups at predefined time points: T0: baseline upon operating room entry; T1: tracheal intubation; T2: skin incision; T3: 30 min post-incision; T4: 60 min post-incision; T5: 90 min post-incision; T6: extubation; T7: post-anesthesia care unit (PACU) discharge.

Statistical analysis and sample size

The sample size was calculated using PASS software, version 15.0 (NCSS, Kaysville, Utah, USA). Based on follow-up data from elderly patients undergoing thoracoscopic lobectomy at our center, the control group exhibited a resting VAS score of 3.25 ± 0.84 immediately post-extubation. Anticipating a 40% reduction in the M1 group, the minimum sample size per group was calculated as 19 cases under the statistical parameters of α = 0.05 and power (1-β) = 0.8. Accounting for a 10% attrition rate, the total required sample size was adjusted to 84 cases.

Continuous variables were expressed as the mean (standard deviation) if the values were normally distributed. If not normally distributed, the median (interquartile range) was used. Normality was assessed using the Kolmogorov-Smirnov test. A one-way ANOVA was used to compare means among the four groups for normally distributed continuous data, while a non-parametric test was used to compare medians for skewed endpoints. A p-value < 0.05 was considered statistically significant. All statistical analyses were performed using SPSS version 25.0 (IBM Corp., Chicago, IL, USA).

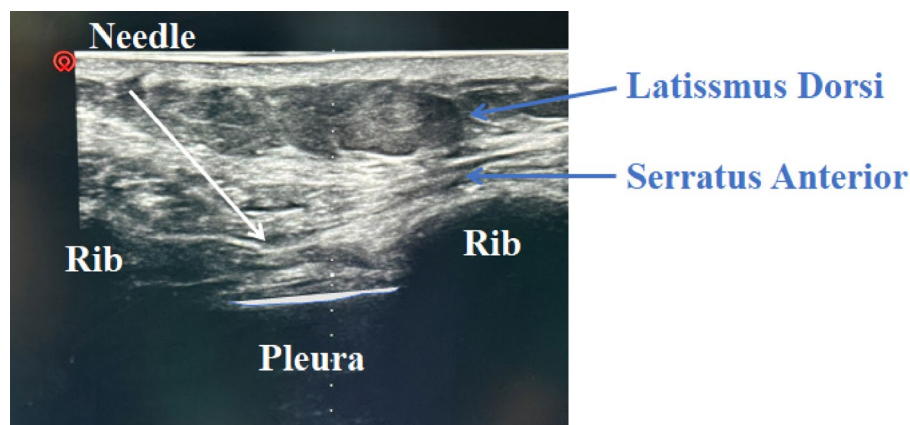


Fig. 1. The white arrow indicates the direction and injection site of the needle tip, with the local anesthetic deposited into the fascial space between the serratus anterior muscle and the ribs or external intercostal muscles.

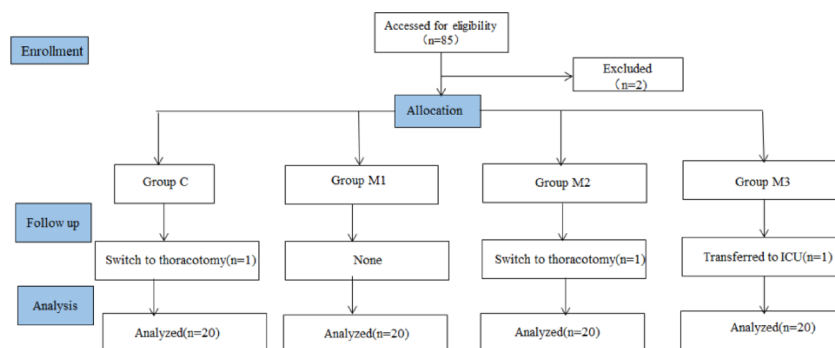


Fig. 2. Consolidated standards of reporting trials flow diagram.

Factor	C	M1	M2	M3	P-value
Age (year)	68.35 ± 1.95	70.85 ± 4.32	70.10 ± 3.21	70.25 ± 4.51	0.166
Sex, n (%)					0.813
Male	10 (50)	11 (55)	8 (40)	10 (50)	
Height (cm)	164.85 ± 6.43	167.00 ± 7.32	170.20 ± 5.93	166.85 ± 8.76	0.138
Weight (kg)	62.95 ± 9.30	65.95 ± 10.08	66.75 ± 9.33	64.40 ± 10.19	0.617
BMI (kg/m ²)	23.09 ± 2.35	23.80 ± 4.23	23.15 ± 3.85	23.29 ± 4.39	0.933
ASA classification, n (%)					0.504
II	19 (95)	18 (90)	20 (100)	18 (90)	
III	1 (5)	2 (10)	0 (0)	2 (10)	
Medical history, n (%)	8 (40)	8 (40)	10 (50)	8 (40)	0.893
Surgery time (min)	114.75 ± 25.20	115.70 ± 22.62	118.60 ± 18.58	120.50 ± 16.49	0.813

Table 1. Anthropometric characteristics. ASA: American Society of Anesthesiologists; BMI: Body Mass Index.

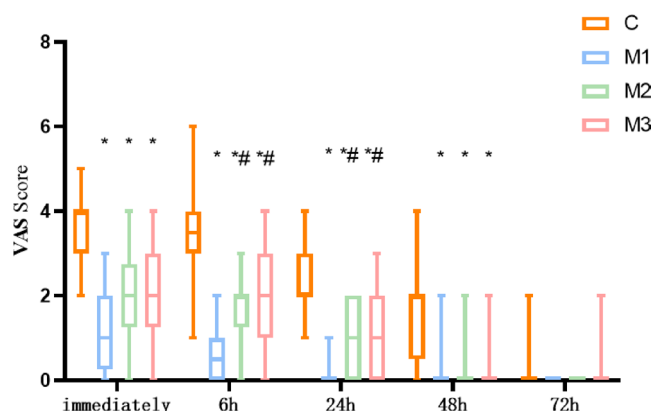


Fig. 3. VAS scores at rest.

Results

85 subjects between August 2024 and November 2024 were enrolled (Fig. 2). Five patients were excluded from the study due to refusal to participate, switch to thoracotomy, and transfer to the Intensive Care Unit (ICU). Anthropometric characteristics were well balanced between the four treatment groups (Table 1).

In the outcome analysis, 20 participants were included in each group, based on the original randomization. No participants crossed over between groups or were excluded after randomization.

VAS scores postoperatively

There were significant differences in VAS scores at rest and on coughing among the four groups immediately after extubation and at 6, 24, 48, 72 h postoperatively (Figs. 3 and 4). In the M1, M2, and M3 groups, the VAS scores at rest were significantly lower than those in the C group immediately after extubation and at 6, 24, and 48 h postoperatively ($p < 0.05$). The VAS scores at rest in the M1 group were significantly lower than those in the

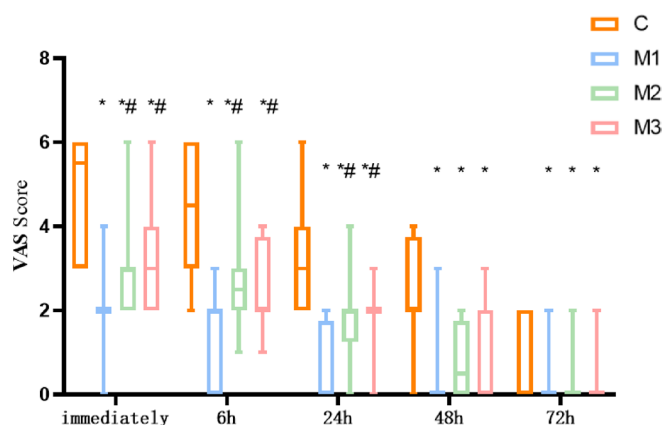


Fig. 4. VAS scores at coughing.

VAS	C	M1	M2	M3	P-value
Immediately	4.00 (3.00,4.00)	1.00 (0.25,2.00)*	2.00 (1.25,2.75)*	2.00 (1.25,3.00)*	<0.01
6 h	3.50 (3.00,4.00)	0.50 (0.00,1.00)*	2.00 (1.25,2.00)*#	2.00 (1.00,3.00)*#	<0.01
24 h	2.00 (2.00,3.00)	0.00 (0.00,0.00)*	1.00 (0.00,2.00)*#	1.00 (0.00,2.00)*#	<0.01
48 h	2.00 (0.50,2.00)	0.00 (0.00,0.00)*	0.00 (0.00,0.00)*	0.00 (0.00,0.00)*	<0.01
72 h	0.00 (0.00,0.00)	0.00 (0.00,0.00)	0.00 (0.00,0.00)	0.00 (0.00,0.00)	0.288

Table 2. VAS scores at rest. VAS, Visual Analogue Scale Scores. *Compared with group C, $P < 0.05$. #Compared with group M1, $P < 0.05$.

VAS	C	M1	M2	M3	P-value
Immediately	5.50 (3.00,6.00)	2.00 (2.00,2.00)*	3.00 (2.00,3.00)*#	3.00 (2.00,4.00)*#	<0.01
6 h	4.50 (3.00,6.00)	2.00 (0.00,2.00)*	2.50 (2.00,3.00)*#	2.00 (2.00,3.75)*#	<0.01
24 h	3.00 (2.00,4.00)	0.00 (0.00,1.75)*	2.00 (1.75,2.00)*#	2.00 (2.00,2.00)*#	<0.01
48 h	2.00 (2.00,3.75)	0.00 (0.00,0.00)*	0.50 (0.00,1.75)*	0.00 (0.00,2.00)*	<0.01
72 h	0.00 (0.00,2.00)	0.00 (0.00,0.00)*	0.00 (0.00,0.00)*	0.00 (0.00,0.00)*	0.009

Table 3. VAS scores at coughing. VAS, Visual Analogue Scale Scores. *Compared with group C, $P < 0.05$. #Compared with group M1, $P < 0.05$.

M2 group and the M3 group at 6 and 24 h postoperatively (Table 2). In the M1, M2, and M3 groups, the VAS scores during coughing were significantly lower than those in the C group immediately after extubation and at 6, 24, and 48 h postoperatively ($p < 0.05$). At 72 h postoperatively, the VAS scores in the M1 and M3 groups were significantly lower than those in the C group. The VAS scores during coughing in the M1 group were significantly lower than those in the M2 and M3 groups immediately after extubation and 48 h after surgery (Table 3).

Horizontal lines indicate overall median for each treatment group; boxes indicate the 25th and 75th percentiles; vertical lines indicate the 5th and 95th percentiles.

Secondary end points

There were minimally significant differences among the four groups ($p < 0.05$) in the intraoperative consumption of remifentanyl, PCIA pressing times, dosage of postoperative analgesics, the dosage of vasoactive drugs, the incidence of postoperative adverse events (Table 4).

Intraoperative monitoring parameters

At the T1 time point, the PTi monitoring results in the M1 and M3 groups showed statistically significant differences compared with the C group ($P < 0.05$). At the T7 time point, statistically significant differences were observed in the PTi monitoring results of the M1, M2, and M3 groups compared with the C group ($P < 0.05$). At all other time points, no statistically significant differences were found among the four groups ($P > 0.05$). (Table 5).

The WLi monitoring results demonstrated no statistically significant differences among the four groups at any time points ($P > 0.05$). (Table 6)

	C	M1	M2	M3	P-value
Remifentanyl dosage ($\mu\text{g}/\text{kg}\cdot\text{min}$)	0.23 ± 0.03	$0.18 \pm 0.02^*$	$0.19 \pm 0.02^*$	$0.18 \pm 0.02^*$	< 0.01
Rescue analgesic dosage (mg)	3.00 (0.00,3.00)	0.00 (0.00,0.00)*	0.00 (0.00,0.00)*	0.00 (0.00,1.00)*	< 0.01
Number of presses on PCIA	2.50 (2.00,3.00)	0.00 (0.00,0.00)*	1.00 (0.00,2.00)*	1.00 (0.00,2.00)*	< 0.01
Vasopressor dosage: norepinephrine (mg)	1.07 ± 0.43	$0.59 \pm 0.33^*$	0.70 ± 0.28	$0.66 \pm 0.33^*$	< 0.01
Occurrence of postoperative adverse events, n(%)					0.02
Hypoxia	4 (20)	1 (5)	1 (5)	1 (5)	
Nausea	4 (20)	2 (10)	3 (15)	1 (5)	
Vomiting	2 (10)	0 (0)	2 (10)	0 (0)	
Agitation	4 (20)	1 (5)	1 (5)	3 (15)	
Delirium	1 (5)	0 (0)	1 (5)	0 (0)	

Table 4. Secondary end points. PCIA, Patient-Controlled Intravenous Analgesia. *Compared with group C, $P < 0.05$.

	C	M1	M2	M3	P-value
T0	92.15 ± 3.08	92.00 ± 4.47	91.60 ± 4.72	90.75 ± 4.95	0.74
T1	55.35 ± 4.71	$48.90 \pm 5.81^*$	51.65 ± 6.60	$49.95 \pm 5.85^*$	< 0.01
T2	50.50 ± 3.79	48.80 ± 5.39	49.75 ± 7.01	50.45 ± 7.05	0.79
T3	53.05 ± 4.23	48.65 ± 6.59	49.85 ± 4.78	48.15 ± 5.77	0.16
T4	50.75 ± 6.00	49.25 ± 5.76	49.40 ± 6.48	48.90 ± 5.60	0.78
T5	49.16 ± 7.03	45.60 ± 5.04	48.60 ± 7.16	50.40 ± 6.68	0.13
T6	80.75 ± 6.64	79.85 ± 8.67	81.15 ± 8.19	78.55 ± 8.91	0.75
T7	85.80 ± 4.49	$79.80 \pm 10.63^*$	$78.35 \pm 9.52^*$	$76.55 \pm 9.03^*$	0.01

Table 5. PTI. PTi, Pain Tolerance Index. *Compared with group C, $p < 0.05$.

	C	M1	M2	M3	P-value
T0	90.65 ± 3.20	90.05 ± 3.38	90.70 ± 3.48	89.80 ± 3.35	0.79
T1	56.65 ± 2.98	55.80 ± 3.32	55.60 ± 4.04	56.05 ± 4.42	0.83
T2	49.25 ± 6.01	49.05 ± 6.19	52.15 ± 5.00	48.95 ± 5.77	0.24
T3	50.05 ± 6.20	50.40 ± 6.95	47.85 ± 5.44	48.75 ± 6.37	0.46
T4	50.30 ± 6.62	52.15 ± 6.32	52.95 ± 6.46	48.90 ± 5.89	0.15
T5	49.85 ± 6.75	49.55 ± 6.43	48.40 ± 6.39	50.40 ± 4.25	0.41
T6	70.75 ± 5.12	70.05 ± 2.78	69.25 ± 3.60	72.55 ± 3.23	0.49
T7	85.00 ± 2.99	84.65 ± 2.66	85.65 ± 3.48	84.00 ± 3.03	0.39

Table 6. WLI.

At the T1 time point, both HR and MAP in the M1 and M3 groups were significantly lower than those in the C group ($P < 0.05$). At the T2 and T7 time points, HR and MAP in the M1, M2, and M3 groups showed statistically significant reductions compared to the C group ($P < 0.05$). No statistically significant differences were observed among the four groups at other time points ($P > 0.05$). (Figures 5 and 6)

There were no protocol deviations or adverse events related to the block.

Discussion

This randomized, double-blind study demonstrated that SAPB combined with oxycodone for multimodal preventive analgesia reduced the postoperative pain in patients.

Multimodal preventive analgesia, as a novel analgesic strategy, achieves multi-target and multi-pathway pain relief by inhibiting central and peripheral sensitization while precisely modulating perioperative nociceptive signaling, thereby reducing the risks of both acute and chronic postoperative pain¹¹. First described by Blanco et al.¹⁰ in 2013, SAPB effectively alleviates thoracic somatic and neuropathic pain by blocking the lateral cutaneous branches of the intercostal nerves (T2–T9)¹². A systematic review encompassing 38 randomized controlled trials (2,224 patients) demonstrated that SAPB and thoracic paravertebral block (TPVB) provided superior pain reduction compared to other regional techniques—including erector spinae plane block (ESPB), local infiltration analgesia, and intercostal nerve block—as evidenced by lower VAS scores at 48 h postoperatively in patients undergoing video-assisted thoracic surgery (VATS)¹³. Wang et al.¹⁴ further confirmed non-inferiority

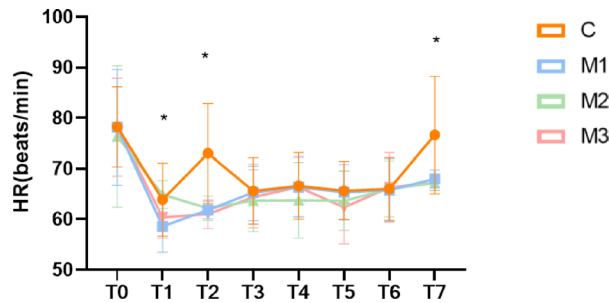


Fig. 5. Intergroup comparison of HR dynamics at predefined intraoperative intervals.

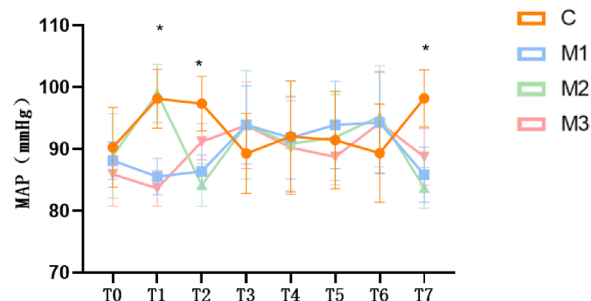


Fig. 6. Intergroup comparison of MAP dynamics at predefined intraoperative intervals.

of SAPB to TPVB in managing post-lobectomy VATS pain. Visceral pain, a predominant contributor to acute postoperative pain in endoscopic surgery, relies on κ -opioid receptor-mediated spinothalamic pathway signaling¹⁵. Oxycodone, a dual μ - and κ -opioid receptor agonist, not only mitigates visceral pain and pleuritic discomfort induced by closed thoracic drainage tubes (CTDTs) via κ -receptor activation¹⁶ but also attenuates inflammatory pain by downregulating pro-inflammatory cytokines such as TNF- α and IL-6^{17–19}. Notably, oxycodone exhibits a lower incidence of opioid-related adverse effects (particularly respiratory depression) and suppresses remifentanyl-induced hyperalgesia by competitively antagonizing κ -receptors against dynorphin—a key mediator of remifentanyl-associated hyperalgesia²⁰. Clinical evidence indicates that 0.1 mg/kg oxycodone, compared to sufentanil, significantly reduces visceral pain intensity at 2 and 4 h postoperatively while maintaining lower systemic inflammatory markers^{21,22}.

This study adopts a multimodal preventive analgesia protocol combining SAPB with oxycodone to achieve comprehensive coverage across four pain pathways: somatic, visceral, inflammatory, and neuropathic. This approach aims to minimize perioperative pain intensity and opioid consumption. Notably, preventive analgesia differs conceptually from preemptive analgesia. The latter emphasizes timing of intervention, positing that preoperative analgesic administration yields superior acute postoperative pain control compared to postoperative administration, based on the rationale of blocking nociceptive processes prior to central sensitization²³. However, animal incisional pain models demonstrate that single-dose preoperative analgesia (whether systemic or neuraxial) only transiently suppresses postoperative pain behaviors until pharmacological effects subside, after which wound nociception re-triggers central sensitization²⁴. Clinical trials corroborate these findings²⁵. Consequently, the critical determinant for preventing central sensitization lies not in intervention timing but in the duration and efficacy of perioperative nociceptive blockade. Current consensus holds that complete and continuous blockade of all surgical trauma-related nociceptive inputs is essential to prevent central sensitization. Thus, this study selects SAPB (with a duration covering the entire VATS lobectomy timeframe) combined with oxycodone as the analgesic regimen to ensure uninterrupted nociceptive blockade.

The results of this study showed that the resting-state VAS score of the M1 group immediately after tracheal extubation, 1.00 (0.25, 2.00), was significantly lower than that of the C group, 4.00 (3.00, 4.00). The cough-state VAS score of the M1 group, 2.00 (2.00, 2.00), was significantly lower than those of the C group, 5.50 (3.00, 6.00), the M2 group, 3.00 (2.00, 3.00), and the M3 group, 3.00 (2.00, 4.00). These findings not only met the minimal clinically important difference (MCID) threshold proposed by Myles et al.²⁶ (a reduction of 10 mm on a 100 mm VAS scale) but also translated clinically relevant improvements by shifting pain intensity from moderate (VAS 4–6) to mild (VAS 1–3), underscoring the efficacy of the multimodal preventive analgesia strategy combining serratus anterior plane block (SAPB) with oxycodone in alleviating acute pain among elderly patients undergoing video-assisted thoracoscopic lobectomy. Although resting VAS scores in the M1 group were numerically lower than those in the M2 and M3 groups, the differences lacked statistical significance, likely attributable to the limited discriminatory capacity of the VAS scale in low-intensity pain ranges (0–3), where patients tend to report integer scores (e.g., 1 or 2). A median intergroup difference of ≤ 1 point would require substantially larger sample sizes to achieve 80% statistical power.

Notably, while the anesthetic protocol involved multi-phase opioid administration, pharmacokinetic analysis suggests minimal confounding effects on the primary outcome (immediate post-extubation VAS scores). Alfentanil, an ultrashort-acting μ -opioid receptor agonist, exhibits an elimination half-life of 145.333 ± 48.915 min in elderly patients. Based on its plasma concentration-time profile modeled by the function $C = 147.16e^{-0.0633t} + 51.55e^{-0.0046t}$, plasma concentrations at extubation (150 min) fell below 50% of the effective analgesic threshold (40–80 ng/mL), indicating negligible residual analgesic effects. Furthermore, postoperative hydromorphone continuous infusion had not yet reached the minimum effective concentration at the time of extubation, ensuring unbiased pain assessment.

The analgesic efficacy observed in this study aligns with but also diverges from prior literature. Chen et al.²⁷ reported postoperative 6-hour VAS scores of 3.00 (3.00, 3.00) in PCIA group (unspecified regimen) versus 3.00 (2.00, 3.00) in a SAPB-PCIA combination group, consistent with our control and SAPB-only groups. However, their SAPB regimen (30 mL 0.375% ropivacaine + 1 mg hydromorphone) yielded higher 6-hour VAS scores [2.00 (2.00, 2.00)] compared to our SAPB-oxycodone (0.1 mg/kg) group [0.50 (0.00, 1.00)], likely attributable to oxycodone's superior κ -opioid receptor affinity and visceral pain modulation synergizing with SAPB's somatic blockade.

Of particular interest, while prior studies reported a SAPB duration of 6.43 h (386 min)¹⁰ and an oxycodone half-life of 3.5 h (duration: 4 h), our trial demonstrated statistically significant VAS reductions in the intervention group persisting up to 72 h postoperatively. This prolonged efficacy suggests synergistic suppression of peripheral and central sensitization rather than mere pharmacodynamic effects. Clarke et al.²⁸ proposed that analgesic interventions exceeding 5.5 drug half-lives in duration qualify as preventive analgesia. Our observed 72-hour efficacy (equivalent to 20.5 oxycodone half-lives) definitively meets this criterion, validating the sustained benefits of multimodal preventive strategies.

Moreover, compared to the C group, the M1, M2, and M3 groups exhibited significant reductions in intraoperative remifentanyl consumption, postoperative rescue analgesia demand, and total perioperative opioid requirements. Consequently, opioid-related adverse events (hypoxia, nausea, vomiting) and postoperative delirium incidence were markedly decreased. These outcomes underscore the clinical value of preventive analgesia in optimizing perioperative pain management: by attenuating central and peripheral sensitization, analgesic demands and associated complications are reduced, ultimately enhancing recovery quality.

Currently, the monitoring of analgesic levels during general anesthesia predominantly relies on the subjective experience of anesthesiologists, who infer analgesic effects through sedation monitoring indicators combined with vital sign parameters (e.g., HR, blood pressure). However, this approach has significant limitations. Particularly in elderly and critically ill patients with impaired autonomic nervous regulation and blunted cardiovascular responses, relying solely on blood pressure fluctuations to assess analgesic needs can result in high misjudgment rates²⁹.

In addition to these traditional methods, several novel pain monitoring tools have emerged in recent years, including the Pain Threshold Index (PTi), Surgical Pleth Index (SPI), and Analgesia Nociception Index (ANI). Although SPI and ANI show potential in guiding precise opioid administration (e.g., reducing remifentanyl dosage by 15–20%) and optimizing hemodynamic management, their reliability remains affected by patient population variability, intraoperative variables, and drug metabolism interference³⁰.

PTi is a data-derived metric based on wavelet index electroencephalogram (EEG) monitoring technology. It reflects the pain intensity of anesthetized patients with high diagnostic accuracy, making it a reliable pain monitoring indicator^{31,32}. The PTi ranges from 0 to 100, with higher scores indicating lower pain tolerance. Researchers recommend maintaining PTi within an optimal range of 40–60 during general anesthesia, applicable to all surgical patients regardless of age. Wang et al.³³ found that PTi measured at the end of surgery had superior accuracy in predicting postoperative pain compared to SPI. However, PTi has limitations: its accuracy may be compromised in patients with conditions or medications affecting EEG activity.

The innovation of this study lies in the following aspects: it is the first to combine serratus anterior plane block (SAPB) with oxycodone for preventive analgesia in video-assisted thoracoscopic lobectomy, establishing a comprehensive and continuous analgesic protocol; secondly, this study proposes a personalized intervention protocol specifically tailored for elderly patients; additionally, it introduces dynamic intraoperative monitoring of the Pain Threshold Index (PTi), enabling precision medication through quantitative assessment of nociceptive responses, significantly enhancing the accuracy of analgesia and data reliability.

This study also has several limitations: first, the sample size is relatively small; second, as a single-center study, the representativeness and applicability of the data may be limited—future research should validate the findings through multicenter, large-scale studies involving diverse populations to improve the generalizability and reliability of the conclusions; finally, the short follow-up period did not allow for tracking the incidence of chronic pain or evaluating patients' postoperative recovery quality, thereby hindering a comprehensive assessment of the advantages of the multimodal preventive analgesia strategy. Future studies should include longer follow-up periods and incorporate evaluations of postoperative recovery quality to further verify the impact of this analgesic strategy on patient recovery and chronic pain incidence.

Future research should focus on conducting large-scale, multicenter prospective randomized controlled trials to validate the efficacy of the multimodal preventive analgesia strategy. Comparative studies of various multimodal preventive analgesia approaches are also needed to further investigate the analgesic effects of specific interventions.

Conclusions

SAPB combined with oxycodone for multimodal preventive analgesia can reduce perioperative pain in elderly patients undergoing thoracoscopic lobectomy, decrease intraoperative and postoperative opioid consumption,

and accelerate their recovery. In future research, large-scale prospective randomized controlled trials are required to compare the efficacy of multimodal preventive analgesia.

Data availability

Data is provided within the supplementary information files.

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Author contributions

HYR and TDD conceptualized and designed the project, including the data collection procedures, and drafted the initial manuscript. QFF and SXH were responsible for performing the nerve block procedure. GZW was responsible for collecting relevant data during and after surgery. LZM conducted the data analyses. All authors contributed to revising the manuscript and approved the final version.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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